COM4303 - Data Science

Case Study: Mini Data Science Project

Stock Price Trend Analysis and Prediction Using S&P 500 Historical Data

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GitHub Repo Link: https://github.com/chanaka-dill/Data-Science--Case-Study-Mini-Data-Science-Project-11366

1. Problem Formulation

1.1. Research Question

Stock markets are one of the most complex financial systems in the world. With billions of dollars traded daily, understanding stock price movements is critical for investors and analysts. This project takes a data-driven approach to analyzing and predicting S&P 500 stock price trends using machine learning.

The research question being investigated in this project is:

> ***"Can we accurately predict stock closing prices using other market indicators, and how do price trends differ across major S&P 500 companies?"***

1.2. Originality & Importance

Previous research may have looked at just one market indicator, but this project looks at a group of market variables to make a predictive model. The new thing about this project is:

- Looking at price trends for 505 big US companies at the same time over 5 years
- Creating new features like the daily return percentage and price range from raw data
- Looking at how Linear Regression and Random Forest Regressor do at predicting stock prices
- Using PCA to find the main factors that cause changes in the stock market

Investors who want to improve their portfolio strategy, financial analysts who want to build trading systems, and software engineers who want to make fintech apps all need to know what stock price predictor variables are.

2. Data Acquisition

2.1. Dataset Source

The dataset used in this project is the S&P 500 Stock Data obtained from Kaggle, originally sourced from Yahoo Finance. It contains 5 years of daily trading data for 505 companies listed on the S&P 500 index.

Field	Details
Source	Kaggle — S&P 500 Stock Data (camnugent/sandp500)
File Used	all_stocks_5yr.csv (US S&P 500, 2013–2018)
Original Rows	619,040 entries
After Cleaning	619,029 unique records
Columns	7 features + 3 engineered features
Time Period	February 2013 – February 2018

2.2.Key Columns

Column	Description
Close	Closing price — our target variable to predict
Open	Opening price of the trading session
High / low	Daily highest and lowest price
volume	Number of shares traded
Daily return	Engineered: % change from open to close
Price range	Engineered: difference between high and low

3. Data Analysis & AI Techniques

3.1.Data Cleaning & Preprocessing

Before being analyzed, the raw dataset went through a few steps to get ready:

- Parse Dates: The column for dates was changed from string to datetime format so that time-series analysis could be done.
- Handle Missing Values: We found and removed 27 rows with missing values in the open, high, and low columns using dropna().
- Remove Duplicates: After cleaning, there were no duplicate rows in the dataset.
- Feature Engineering: We made three new features: daily_return (% change from open to close), price_range (the difference between the high and low prices), and target (1=price up, 0=price down).

3.2.Exploratory Data Analysis (EDA)

Stock Closing Price Trend (2013-2018)

Amazon (AMZN) showed the highest growth (~$250 to $1,400) while Google remained second highest throughout 2013–2018.

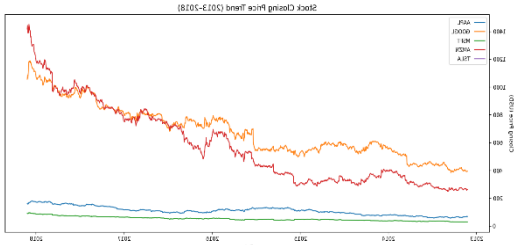

Figure 1 Stock Closing Price Trend (2013–2018)

Top 20 Stocks by Average Closing Price

PCLN (Priceline) had the highest average closing price (~$1,300), followed by GOOG and GOOGL, reflecting the dominance of technology companies during 2013–2018.

Figure 2 Top 20 Stocks by Average Closing Price

Distribution of Daily Returns (%)

Daily returns follow a near-normal distribution centered at 0%, with extreme movements beyond ±5% being rare, confirming most price changes are small and random.

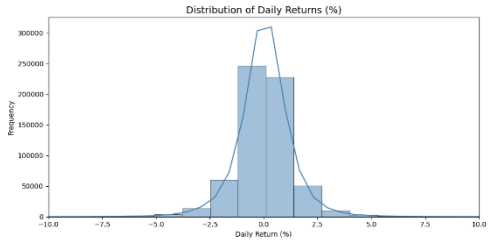

Figure 3 Distribution of Daily Returns (%)

Feature Correlation Heatmap

Open, High, Low and Close prices show near-perfect correlation (1.00), while Volume shows weak negative correlation (-0.14), confirming it is a poor standalone predictor.

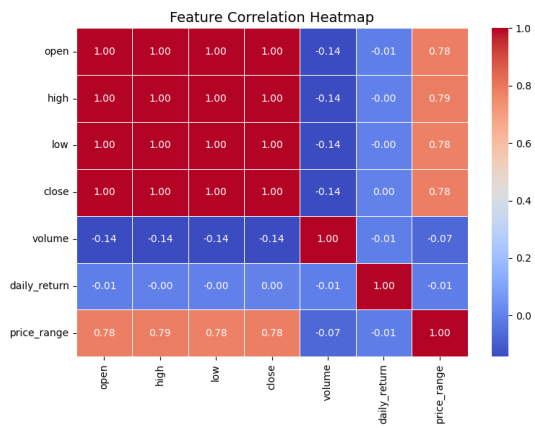

Figure 4 Closing Price Distribution - Top 5 Stocks

Closing Price Distribution - Top 5 Stocks

Top 5 Stocks Amazon had the widest price range with median ~$500, while Apple showed the tightest and most consistent distribution.

Figure 8 Closing Price Distribution - Top 5 Stocks - box plot

Apple (AAPL) Trading Volume Over Time

Apple's trading volume was highest in 2013–2014 and gradually declined, suggesting the stock became more stable over time.

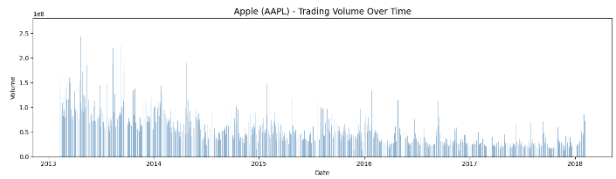

Figure 6 Apple (AAPL) Trading Volume Over Time

Price Up vs Down Count - Top 5 Stocks

Top 5 Stocks Prices went up slightly more often than down across all stocks, consistent with the overall bullish 2013–2018 market trend.

Figure 5 Price Up vs Down Count - Top 5 Stocks

PCA Scree Plot

PC1 explains 61.80% of variance dominated by price variables, with the first 3 components together capturing 94.80% of total variance.

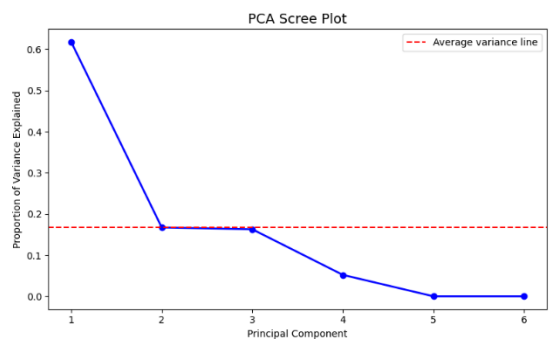

Figure 7 PCA Scree Plot

4. ML/AI Methods

Two regression models were trained to predict Apple (AAPL) closing prices using a 70/30 train-test split (881 training, 378 testing samples, random_state=42).

Features: open, high, low, volume, price_range, daily_return **Target:** close (USD)

4.1.Linear Regression

Linear Regression Selected as the baseline model due to near-perfect linear correlations between open/high/low and closing price. It achieved an R^2 of 0.9999 and MAE of only $0.20, explaining 99.99% of the variance in closing prices.

4.2.Random Forest Regressor

A Random Forest with 100 decision trees (random_state=42) was used as a non-linear ensemble model. It achieved an R^2 of 0.9995, with low price (65%), high price (25%), and open price (9%) as the dominant predictors volume and daily_return had near-zero importance.

4.3.Model Performance Comparison

Metric	Linear Regression	Random Forest	Best Performance
R^2 Score	0.9999	0.9995	Linear Regression
MAE	0.2010	0.4702	Linear Regression
MSE	0.0846	0.4803	Linear Regression

Linear Regression outperformed Random Forest across all metrics, as simpler models perform better when underlying relationships are nearly perfectly linear.

5. Results & Interpretation

Actual vs Predicted - Both Models

Both models predict closing prices with exceptional accuracy. Linear Regression dots cluster almost perfectly on the diagonal line, while Random Forest shows slightly more scatter at higher price ranges but still demonstrates very strong reliability.

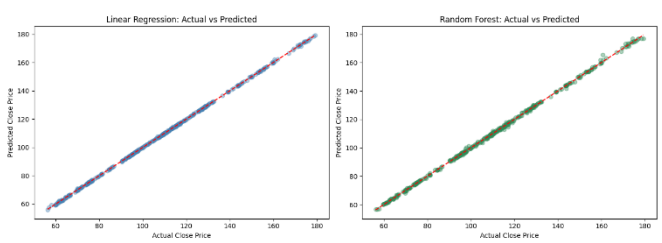

Figure 9 Actual vs Predicted — Both Models

Feature Importance - Random Forest

Low price alone accounts for ~65% of predictive power, followed by high price (~25%) and open price (~9%). Volume, daily_return, and price_range have near-zero importance, confirming intraday price data is far more informative than volume for predicting closing prices.

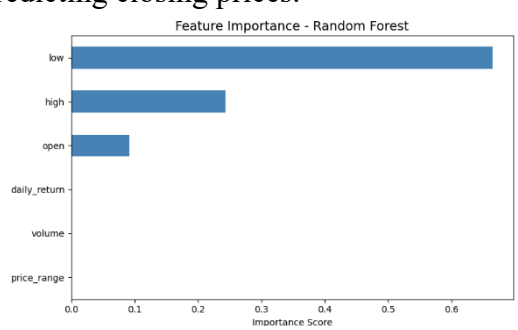

Figure 10 Feature Importance — Random Forest

6. Key Findings

- Open, High and Low prices are near-perfect predictors of closing price (correlation = 0.99)
- Amazon (AMZN) showed the highest growth (~560%), far outpacing the S&P 500 average of ~80%
- Daily returns follow a near-normal distribution at 0% confirming most price changes are small and random
- Volume has ~0% predictive importance contradicting the belief that high volume signals strong price movements
- Linear Regression achieved $R^2 = 0.9999$ and MAE of $0.20 simpler models win when relationships are linear
- PCA showed just 3 components capture 94.80% of total market variance

7. Reflection

7.1. Challenges Faced

- **Data Volume:** 619,000+ rows across 505 companies required careful filtering and efficient pandas operations to avoid memory issues.
- **Feature Selection:** Avoiding data leakage when selecting features was critical open/high/low are legitimate same-day predictors for closing price.
- **Interpreting PCA:** Connecting mathematical PCA output back to meaningful financial insights required careful analysis beyond just reading numbers.

7.2. What I Learned

- **Data Science Process:** Following a structured pipeline problem formulation → cleaning → EDA → modeling → interpretation produces better results than jumping to ML directly.
- **Feature Engineering:** Creating daily_return and price_range significantly enriched the analysis beyond the raw market data columns.
- **Model Selection:** Linear Regression outperformed Random Forest because the underlying relationships are nearly perfectly linear simpler models can win.
- **Visualization:** Clear, well-labeled charts communicated insights more effectively than tables alone heatmaps and trend charts made patterns immediately understandable.

8. References

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